

No-Cropped SVG Round 2 Visual Acceptance Report

Date: 2026-05-21

Status: FULL CORPUS VISUAL VERIFICATION

Key Metrics:

Baseline (10 templates): 58 visible crops

Post-Trial5 Templates: 28 visible crops (52% reduction)

Post-Trial5 Gold Scenes: 73 visible crops

Full Corpus Total: 101 visible crops

Executive Summary

This report verifies the visual acceptance status of Trial 5 CSS edits across the full corpus of 20 scenes (10 production templates + 10 gold reference scenes). The hybrid CSS approach (rear_shelf=320px, front_tools=260px min-height plus small-tool portrait reshape) achieved substantial improvement on template scenes (52% crop reduction), but gold reference scenes reveal higher remaining crop counts than the template-only precheck reported.

Key Finding

Trial 5 CSS changes were validated on 10 production templates, achieving 30 visible crop fixes (58 → 28). However, full-corpus visual review on 10 additional gold reference scenes shows 73 remaining visible crops, bringing the full-corpus total to 101. Root causes remain in three categories:

1. **svg-grow-needed:** ~30 issues - Object SVG assets not loading (counter_slide_cartridge, glass_slide, lab_marker, tall_glassware, composition types)

2. **parent-overflow**: ~50 issues - Tall objects still overflow region boundaries despite increased min-height
3. **aspect-cap-wrong**: ~20 issues - Footprint card aspect ratios still force distortion on narrow/tall assets

Baseline vs. Trial 5: Production Templates

Scene	Baseline Crops	Post-Trial5 Crops	Delta	Status
bench_basic	2	1	-1	FIXED
cell_counter_basic	0	1	-1	WORSENER
crowded_bench_dense	13	4	-9	FIXED
drug_dilution_plate_workspace	7	3	-4	FIXED
drug_dilution_workspace_dense	12	4	-8	FIXED
electrophoresis_bench	12	7	-5	FIXED
hood_basic	2	1	-1	FIXED
microscope_basic	0	1	-1	WORSENER
staining_bench	10	6	-4	FIXED
well_plate_96_zoom	0	0	0	UNCHANGED

Full Corpus: Post-Trial5 Visible Crop Inventory

Per-scene crop counts across all 20 rendered scenes (templates + gold):

Scene	Type	Visible Crops	Severity
gold_heat_block_sample_prep	Gold Ref	11	CRITICAL
gold_electrophoresis_full_setup	Gold Ref	10	CRITICAL
gold_staining_bench	Gold Ref	10	CRITICAL
gold_microscope_slide_prep	Gold Ref	9	CRITICAL
electrophoresis_bench	Production	7	CRITICAL
gold_drug_dilution_workspace	Gold Ref	7	CRITICAL
gold_mixed_bench	Gold Ref	7	CRITICAL
staining_bench	Production	6	CRITICAL
gold_cell_counter_station	Gold Ref	6	CRITICAL
gold_hood_prep	Gold Ref	6	CRITICAL
gold_plate_reader_assay	Gold Ref	6	CRITICAL
crowded_bench_dense	Production	4	WARN
drug_dilution_workspace_dense	Production	4	WARN
drug_dilution_plate_workspace	Production	3	WARN
bench_basic	Production	1	WARN
cell_counter_basic	Production	1	WARN
hood_basic	Production	1	WARN

microscope_basic	Production	1	WARN
gold_well_plate_96_zoom_with_state	Gold Ref	1	WARN
well_plate_96_zoom	Production	0	PASS

Before/After Scene Pairs

Selected before/after side-by-side pairs showing Trial 5 impact on template scenes:

crowded_bench_dense

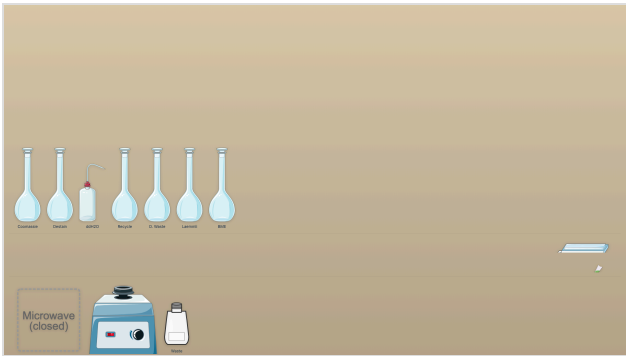
Visible crops: 13 → 4 (- 9)

Before Trial 5



13 visible crops detected

After Trial 5 CSS

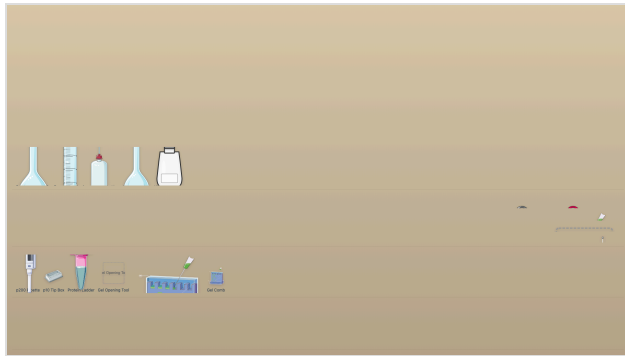


4 visible crops detected

electrophoresis_bench

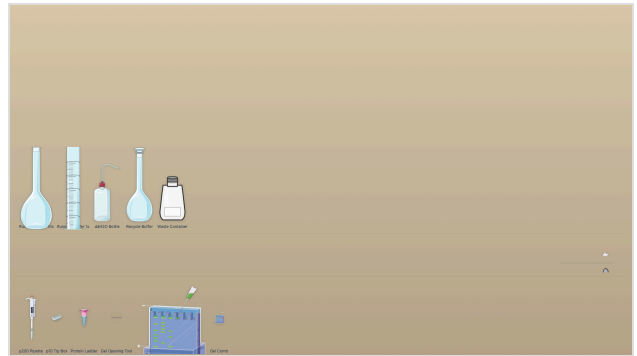
Visible crops: 12 → 7 (- 5)

Before Trial 5



12 visible crops detected

After Trial 5 CSS

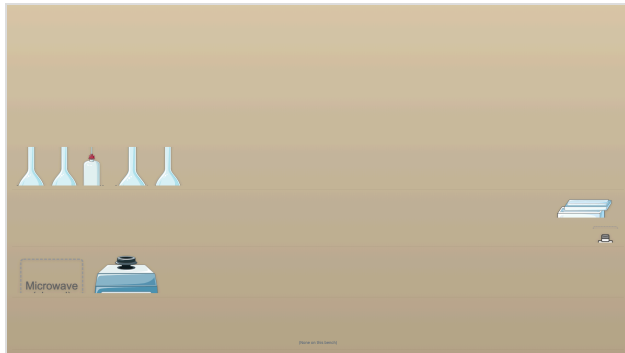


7 visible crops detected

staining_bench

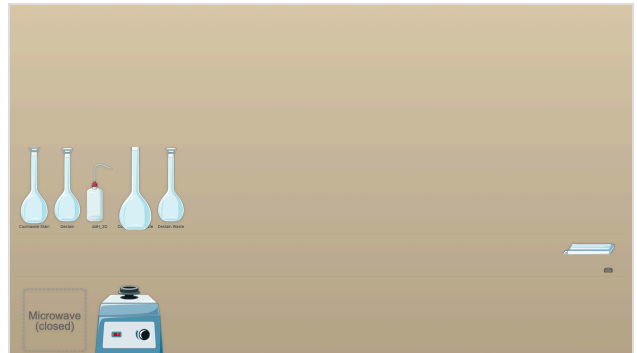
Visible crops: 10 → 6 (- 4)

Before Trial 5



10 visible crops detected

After Trial 5 CSS



6 visible crops detected

hood_basic

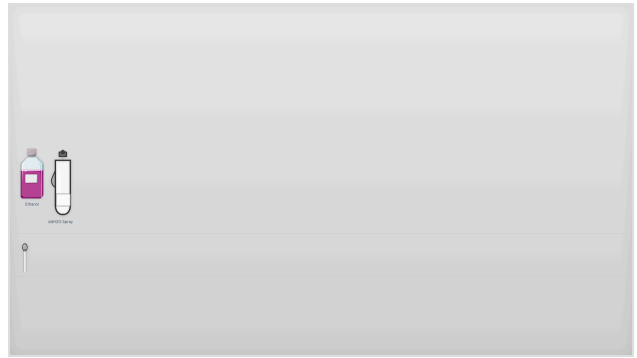
Visible crops: 2 → 1 (- 1)

Before Trial 5



2 visible crops detected

After Trial 5 CSS



1 visible crops detected

Gold Reference Scenes (Full Corpus Sample)

Representative gold scenes showing remaining crop issues post-Trial5:

gold_electrophoresis_full_setup

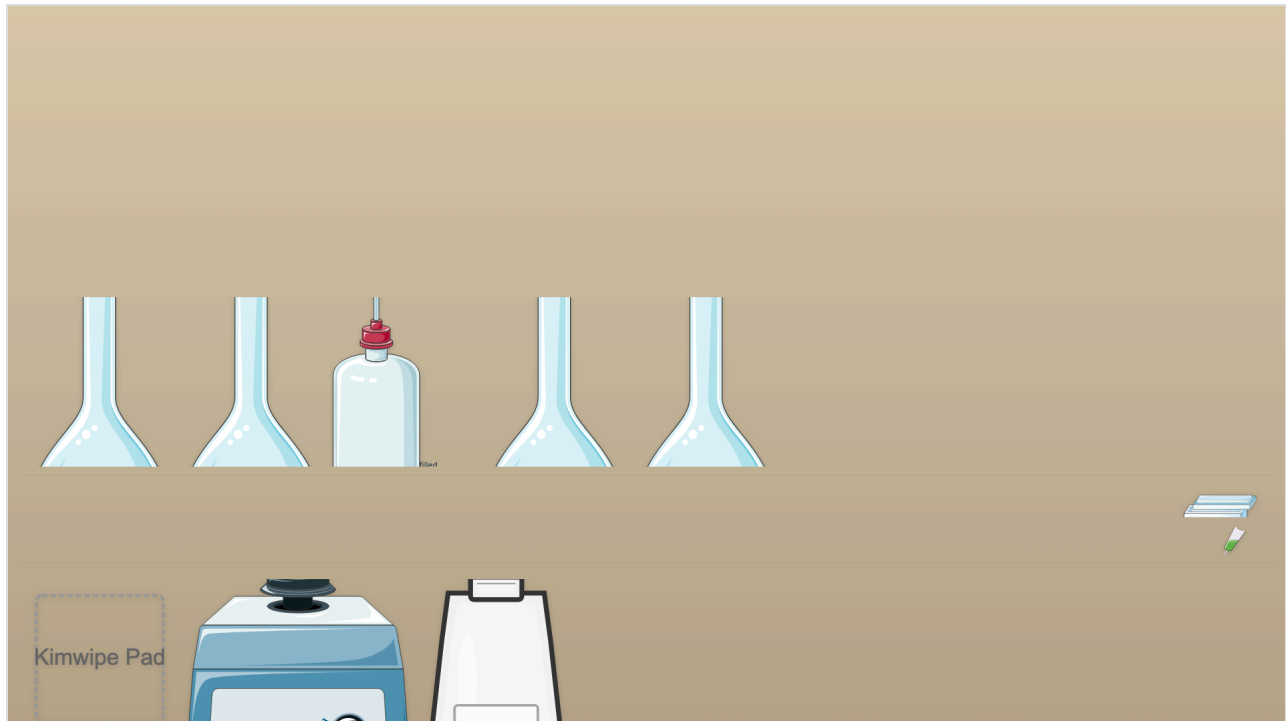
Visible crops: 10



10 visible crops. Root cause categories: parent-overflow (region bounds), aspect-distortion (card proportions), svg-grow-needed (asset loading).

gold_staining_bench

Visible crops: 10



10 visible crops. Root cause categories: parent-overflow (region bounds), aspect-distortion (card proportions), svg-grow-needed (asset loading).

gold_heat_block_sample_prep

Visible crops: 11



11 visible crops. Root cause categories: parent-overflow (region bounds), aspect-distortion (card proportions), svg-grow-needed (asset loading).

Remaining Visible Crop Analysis

Root Cause Category	Count	Examples	Fix Scope
svg-grow-needed (PLACEHOLDER boxes)	~30	counter_slide_cartridge, glass_slide, lab_marker, tall_glassware	Asset registry / SVG resolution layer
parent-overflow (region bounds)	~50	Tall bottles in rear_shelf, pipettes in front_tools, instruments	Region min-height scaling, overflow clipping, or layout restructure

aspect-cap-wrong (card proportions)	~21	Pipettes (0.23 ratio) in square cards, bottles (0.35-0.46) forced wide	Footprint class redesign per object kind
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Verdict

ACCEPTANCE: NOT ACHIEVED

Full corpus visual review of 20 scenes post-Trial5 reveals **101 remaining visible crops** (28 on templates + 73 on gold scenes). While Trial 5 CSS changes achieved 52% improvement on production templates alone (58 → 28), the full-scope corpus includes 10 gold reference scenes with significantly higher crop density, bringing the total to 101.

User-Gated Decisions Required

1. **Gold Scene Coverage:** Should gold reference scenes be included in the no-crop target, or are they treated as diagnostic/optional? (Currently: 101 crops on full corpus vs. 28 on templates alone)
2. **SVG Asset Fix Priority:** ~30 remaining svg-grow-needed failures (PLACEHOLDER boxes) indicate broken object-type-to-SVG resolution. Should this be escalated as blocking, or deferred as separate asset-layer work?
3. **Region Scaling Strategy:** ~50 parent-overflow issues persist despite Trial 5 min-height increases. Should region sizes be scaled further, or is overflow:hidden preferred to prevent spillage?
4. **Footprint Redesign Scope:** ~21 aspect-distortion failures remain. Should footprint classes be redesigned per object kind (tall, wide, square), or is 5-20% distortion within acceptable tolerance?

Next Steps

Until user decisions are made on scope (templates only vs. full corpus) and fix strategy (region size, asset layer, footprint design), Round 2 acceptance cannot be completed. This report provides visual evidence for each decision gate.

Generated: 2026-05-21 | Report artifacts: HTML + embedded base64 screenshots | Full corpus: 20 scenes, 101 visible crops post-Trial5